

QUANTITATIVE CHEMISTRY – GAS VOLUMES

KEY VOCABULARY

Molar gas volume: the volume of one mole of any gas at rtp = 24 dm³.

rtp: room temperature and pressure (20°C, 1 atm).

Avogadro's law: equal volumes of gases have equal moles (same conditions).

dm³: 1 dm³ = 1000 cm³ = 1 litre.

COMMON MISCONCEPTIONS

~~Heavier gases take up more volume.~~

✓ At rtp, one mole of ANY gas is 24 dm³.

~~24 dm³ works at any temperature.~~

✓ It only applies at room temperature and pressure.

~~cm³ and dm³ are the same.~~

✓ Divide cm³ by 1000 to get dm³.

GAS VOLUME AT RTP

$$\text{moles} = \frac{\text{volume (dm}^3\text{)}}{24}$$

CONVERSION REMINDER

1 mole of gas = 24 dm³ at rtp

1 VOLUME → MOLES

Calculate the moles in 48 dm³ of CO₂ at rtp. [2 marks]

Answer: _____ mol

2 MOLES → VOLUME

Calculate the volume of 0.5 mol of O₂ at rtp. [2 marks]

Answer: _____ dm³

3 FROM MASS

Calculate the volume of 8 g of CH₄ at rtp. (M_r = 16) [3 marks]

Answer: _____ dm³

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

At rtp, one mole of any gas occupies _____ dm³, so volume ÷ 24 gives _____.

24

1000

moles

mass

6 REACTION VOLUME

2 mol of gas is produced. Calculate the total gas volume at rtp. [1 mark]

Answer: _____ dm³

TIP: At rtp: moles = volume (dm³) ÷ 24. Convert cm³ to dm³ first.

✓ Think: Why does the type of gas not change the volume of one mole at rtp?